CNAME vs A records

For Benjamin Duivesteyn, deAutomation — 6 November 2026

Overview

An **A record** answers one question: which IPv4 address does this name point at? A **CNAME** answers a different one: this name is an alias for another name, so go and look that name up. Everything else — who maintains the address, what else can live at the name, how a change propagates — follows from that single difference.

The distinction stopped being theoretical on 6 November, when `data.deautomation.co` moved into a Google Cloud DNS zone. The apex stayed an A record, the product hosts became CNAMEs, and at Namecheap only NS records remain. That is now the standard shape for every host we sell.

Details

The difference in one table

Question	A record	CNAME
What it stores	An IPv4 address (`216.239.32.21`)	Another DNS name (`ghs.googlehosted.com.`)
Who owns the address	Us — we edit it when the target changes	The target's owner — we never see the address
Extra lookup at resolve time	None	Yes: the resolver re-queries the target
Allowed at a zone apex	Yes	No (it would collide with the zone's own SOA and NS)
Can other records sit at the same name	Yes (TXT, MX, NS, subdomains)	No — a CNAME must be the only record at its name
Can the name have children	Yes (`cwp.data` under `data`)	Not practically — it is an alias, not a branch
Changes when the host moves	We update the record	The target updates its A record; we do nothing

Two related record types matter here. **AAAA** is the same as A for IPv6. **ANAME/ALIAS** is a provider-side trick that behaves like a CNAME at an apex by resolving the target on the provider's side; Google Cloud DNS does not offer it, which is exactly why our apexes are plain A records.

Where each lives in our estate

Name	Type	Value	Why
`data.deautomation.co`	A	`216.239.32.21`, `.34.21`, `.36.21`, `.38.21`	Apex of the delegated zone; must keep the SPF TXT and the product hosts beneath it
`clients.deautomation.co`	A	The same four addresses	Same shape, already delegated to Cloud DNS
`cwp.data.deautomation.co`	CNAME	`ghs.googlehosted.com.`	Product host pointing at Google's frontend; Google owns the addresses
`ppc.data.deautomation.co`	CNAME	`ghs.googlehosted.com.`	Added 6 November for the new PPC site
`rc.clients.deautomation.co`	CNAME	`ghs.googlehosted.com.`	Same pattern under the clients zone
`data.deautomation.co`	NS	`ns-cloud-c1-c4.googledomains.com.`	Delegation: tells resolvers Google is now authoritative
`data.deautomation.co`	TXT	`v=spf1 include:_spf.mailersend.net ~all`	Mail authorisation; cannot sit beside a CNAME

The NS row is the one people miss. Delegation is not an address record at all: it says who to ask next. It is what lets us add `sgp.data` or `ops.data` later without touching Namecheap again.

What a resolver actually does for `ppc.data.deautomation.co`

1. The stub resolver asks the root, then `.com`, then the `deautomation.co` nameservers at Namecheap.
2. Namecheap returns the **NS** set for `data`: `ns-cloud-c1` through `ns-cloud-c4`.
3. The resolver asks Google for `ppc.data`; the zone answers with a **CNAME** to `ghs.googlehosted.com.`
4. The resolver asks for `ghs.googlehosted.com` and gets an **A** record — currently `192.178.187.121`.
5. The browser connects to that address, sends `ppc.data.deautomation.co` in the TLS handshake, and Google's frontend routes it to the `ppc-data` Cloud Run service.

Until the Namecheap delegation exists, step 2 never reaches Google, so `ppc.data` is still NXDOMAIN — the CNAME exists only inside the new zone. That is the one reason the record is not live yet.

Why the apex is an A record

- A zone apex must carry SOA and NS records, and a CNAME cannot coexist with any other record.
- `data` has children (`cwp.data`, `ppc.data`) and a TXT (SPF). A CNAME at `data` would make all of those impossible.
- The four addresses are Google's frontend for the `data-catalogue` mapping. They are stable in practice, and they are ours to keep correct.

Why the product hosts are CNAMEs

- Google owns `ghs.googlehosted.com` and its addresses. A CNAME means Google can move the frontend without us editing anything.
- Each product host is one mapping and one certificate; the CNAME is the only record the mapping needs.
- It keeps the registrar out of routine work: adding a product is a record in `data-zone`, not a Namecheap edit.

Traps worth remembering

- Never put a CNAME at a zone apex. It looks tidy and breaks the zone.
- A CNAME blocks every other record at that name: no TXT, no MX, no NS, no children. Mail and aliases cannot share a name.
- `dig +short TXT cwp.data.deautomation.co` prints the CNAME target, not a TXT answer. Use `dig TXT +noall +answer` when checking mail records.
- A CNAME adds a lookup and a failure point; keep it one hop, not a chain.
- TTL matters on cutover day, not after: the old Namecheap records cached for 1800 seconds, the new zone uses 300.
- Records are edited where they are authoritative. After delegation, a Namecheap A/TXT edit for `data` is a no-op.

What changed on 6 November

A `data-zone` was created in Cloud DNS and populated with the four apex addresses, the SPF TXT and the `cwp.data` CNAME before anything was touched at the registrar, so delegation cannot produce an empty zone. The `ppc.data` CNAME was added to the same zone and the Cloud Run mapping for it was created. The zone was verified directly against `ns-cloud-c1`. The remaining step is the delegation itself, which is a registrar action and cannot be automated with the credentials we hold.

Next steps

1. Replace the four `data` A records and the `data` TXT at Namecheap with four NS records for `ns-cloud-c1-c4.googledomains.com.`
2. Confirm propagation with `dig NS data.deautomation.co` and `dig CNAME ppc.data.deautomation.co`.
3. Allow 15–30 minutes for the `ppc.data` certificate, then check the mapping reports `Ready: True`.
4. Add future product hosts (`sgp.data`, `ops.data`) as records in `data-zone`; the registrar is not involved again.

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